

Topic: CO1

Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability

The security mechanism used to provide the identified security feature.

The security protocol or technology used to implement the mechanism.

Briefly explain how the mechanism/protocol provides the identified security feature.

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Whatsapp	Confidentiality	End-To-End Encryption	Signal Protocol	Messages and call is encrypted on sender side and later decrypted on receiver side .This helps in not access from unauthorized user.
Google Pay	Authentication	Transaction Authentication	UPI Pin	In this user has to provide correct upi pin for successful transaction to complete .This prevents the unauthorized people getting access.
Paypal	Authentication	Two-Factor Authentication	Authenticator App	This add extra steps from getting unauthorized access.Here not only knowing password is not sufficient to get unauthorised access.
GitHub	Authorization	Repository Access Control	Repository Roles/Permission	This gives user full control who can view ,modify ,manage repository.

Instagram	Authentication	Multi-Factor Authentication	Authenticator App	During logging in ti ask various info like username, password, otp, Security code,etc. given by user for preventing from unauthorized access.
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2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Applicati on	CIA Classification	Security Control	Security Mechanism	Protocol/ Technology	Preventio n/ Detection / Recovery	How it works
WhatsApp	Confidentiality	End-to-End Encryption	Cryptograph ic encryption	Signal Protocol	Preventio n	This encrypt message and call from unauthorise d access.
WhatsApp	Availability	Account and device recovery	Account recovery and device managemen t	Whatsapp security mechanis m	Recovery	It helps user to get access back of the account if lost access.
UPI	Confidentiality	Secure communicat ion	Encryption of data in transit	TLS	Preventio n	This protects data between application payment infrastru ctu re while transferring .
UPI	Integrity	Transaction authenticati on	User authorizatio n	UPI PIN	Preventio n	This requires the user authorize the payment before

						transaction is completed.
UPI	Availability	Payment infrastructure	Distributed payment processing	UPI	Availability	It supports continuous payment and provides mechanism to handle failed transaction
Instagram	Confidentiality	Multi-Factor Authentication	Additional authentication factor	2FA / Authenticator Apps	Prevention	This adds extra security like two factor authentication code
Instagram	Integrity	Login monitoring	Security alerts and activity monitoring	Security Checkup / Login Alerts	Detection	This helps users identify unrecognized login attempts and suspicious changes to their account.
Instagram	Availability	Account recovery	Identity and account verification	Account Recovery / Security Checkup	Recovery	This helps user to get back the access back again.
Microsoft Teams	Confidentiality and Integrity	Real-time media protection	Encrypted and authenticated media	S RTP	Prevention	This protects audio and video from unauthorized interception, modification, and replay attacks.
Microsoft Teams	Availability	Cloud resilience	Redundancy and fault tolerance	Microsoft 365	Recovery	This maintains service availability when

						individual cloud infrastructure components fail.
AWS	Confidentiality	Encryption in rest and transit	Data encryption and secure communication	AWS AES,TLS	Prevention	This encrypt message during storage and transferring between user and application
AWS	Integrity	Audit logging	Activity recording	AWS CloudTrail	Detection	It records user and API activity, allowing administrators to identify and investigate unauthorized actions.
AWS	Availability	Backup and replication	Data recovery and redundancy	Amazon S3	Recovery	This maintains additional copies of data that can be used to restore information after failures or attacks.